



PYTHON NOTES — PYTHON BASICS



TOPIC: PYTHON BASICS

Python Basics is the foundation of my programming journey. These are the first concepts I learned and the building blocks I will continue using in later Python topics.

1. WHAT IS PYTHON?

Python is a programming language used to create programs, applications, websites, games, automation tools, data projects, and many other types of software.

Python is the main programming language I am currently learning.

2. USING VS CODE

I use **VS Code** to write and run my Python programs.

My basic workflow is:

1. Open VS Code.
 2. Create a Python file.
 3. Write Python code.
 4. Save the file.
 5. Run the program.
 6. Read the output and fix errors when necessary.
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3. `print()`

The `print()` function displays information on the screen.

Example

```
print('Hello World')
```

Output

```
Hello World
```

We can also print variables:

```
name = 'Christabel'  
print(name)
```

Output:

```
Christabel
```

4. VARIABLES

A variable is a name used to store information or data.

Example

```
name = 'Christabel'  
age = 14
```

Here:

- `name` stores the string `'Christabel'`
- `age` stores the integer `14`

Variables can be used later in a program:

```
name = 'Christabel'  
print(name)
```

5. `input()`

The `input()` function allows the user to enter information into a Python program.

Example

```
name = input('What is your name? ')
```

The user enters a response, and Python stores it in the `name` variable.

Important

Information received using `input()` is treated as a **string** by default.

For example:

```
age = input('How old are you? ')
```

Even if the user types:

```
14
```

Python initially receives it as text.

6. `int()` AND `int(input())`

`int()` converts a value into an **integer** when the value can be converted successfully.

An integer is a whole number without a decimal point.

Example

```
age = int(input('How old are you? '))
```

If the user enters:

```
14
```

Python stores it as the integer `14`.

Another example

```
num1 = int(input('Type a number: '))
num2 = int(input('Type another number: '))

print(num1 + num2)
```

This program receives two whole numbers and adds them.

7. float() AND float(input())

`float()` converts a value into a **floating-point number**, also called a float.

A float is a number that can contain a decimal point.

Example

```
height = float(input('What is your height? '))
```

If the user enters:

```
1.65
```

Python stores it as a float.

Another example

```
price = float(input('What is the price? '))
print(price)
```

If the user enters `12.50`, Python can store it as a decimal number.

Difference between `int()` and `float()`

`int()` → whole numbers

```
age = int(input('How old are you? '))
```

`float()` → numbers that may contain decimals

```
height = float(input('What is your height? '))
```

Both are used to **convert input into numbers**.

8. STRINGS

A string is text.

Strings are usually written inside quotation marks.

Examples

```
name = 'Christabel'  
food = 'Rice'  
hobby = 'Coding'
```

These are strings:

```
'Hello'  
'Python'  
'14'  
'123.5'
```

Even `'14'` and `'123.5'` are strings because they are inside quotation marks.

9. INTEGERS

An integer is a whole number without a decimal point.

Examples

```
age = 14  
score = 100  
number = 25
```

We can use `int(input())` when we want to receive a whole number from the user.

10. FLOATS

A float is a number that can contain a decimal point.

Examples

```
height = 1.65  
price = 12.50  
temperature = 36.5
```

We can use `float(input())` when we want to receive a number that may contain a decimal point.

11. BASIC ARITHMETIC OPERATORS

Python can perform mathematical calculations using arithmetic operators.

Addition +

Adds two numbers.

```
print(5 + 3)
```

Output:

```
8
```

Subtraction -

Subtracts one number from another.

```
print(5 - 3)
```

Output:

```
2
```

Multiplication *

Multiplies two numbers.

```
print(5 * 3)
```

Output:

```
15
```

Division /

Divides one number by another.

```
print(6 / 3)
```

Output:

```
2.0
```

Example using variables

```
num1 = int(input('Type a number: '))  
num2 = int(input('Type another number: '))  
  
print(num1 + num2)  
print(num1 - num2)  
print(num1 * num2)  
print(num1 / num2)
```

12. USING COMMAS IN print()

When printing multiple pieces of information, we can separate them with commas.

Example

```
name = 'Christabel'  
age = 14  
  
print('My name is', name, 'and I am', age, 'years old')
```

The commas allow us to print different pieces of information in one `print()` statement.

Important

This is incorrect:

```
print('My name is', name ; 'I am', age)
```

Use commas instead:

```
print('My name is', name, 'I am', age)
```

13. COMMENTS

Comments are notes written inside code for humans to read.

A single-line comment begins with `#`.

Example

```
# Ask the user for their name  
name = input('What is your name? ')
```

Python does not execute the comment as program instructions.

Comments can help explain what different parts of a program do.

14. COMPARISON OPERATORS

Comparison operators compare values and produce a Boolean result such as `True` or `False`.

Equal to ==

```
print(5 == 5)
```

Output:

```
True
```

Not equal to !=

```
print(5 != 3)
```

Output:

```
True
```

Greater than >

```
print(8 > 5)
```

Output:

```
True
```

Less than <

```
print(3 < 5)
```

Output:

```
True
```

Greater than or equal to >=

```
print(5 >= 5)
```

Output:

True

Less than or equal to <=

```
print(4 <= 5)
```

Output:

True

Note: Comparison operators are introduced here as part of the Python Basics foundation. Their use in actual decisions belongs to the **Decision Making** topic.

15. PYTHON BASICS PRACTICE PROGRAM

Here is a simple program that combines several Python Basics concepts without using the Decision Making topic.

```
name = input('What is your name? ')
age = int(input('Type in your age: '))
hobby = input('What is your favorite hobby? ')
score = int(input('Give me your exam score: '))

print(
    'My name is',
    name,
    ', I am',
    age,
    'years old. I love',
    hobby,
    'and my exam score is',
    score
)
```

Concepts used

- `input()`
- Variables
- Strings
- Integers

- `int()`
- `print()`
- Commas in `print()`

This is a good example of combining the Python Basics concepts I have learned so far.

WHAT I CAN DO NOW

After learning Python Basics, I can:

- ✓ Create and run Python programs in VS Code
 - ✓ Display information with `print()`
 - ✓ Store information in variables
 - ✓ Ask users for information with `input()`
 - ✓ Work with strings
 - ✓ Work with integers
 - ✓ Work with floats
 - ✓ Convert input using `int()`
 - ✓ Convert input using `float()`
 - ✓ Perform basic arithmetic
 - ✓ Use comments
 - ✓ Compare values using comparison operators
 - ✓ Combine several Python Basics concepts into a simple program
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MY PROGRESS

Current Level: Python Basics ✓

Topics Learned

- ✓ What Python is
- ✓ Using VS Code
- ✓ `print()`
- ✓ Variables
- ✓ `input()`
- ✓ `int()`
- ✓ `float()`
- ✓ Strings
- ✓ Integers
- ✓ Floats
- ✓ Basic arithmetic
- ✓ Comments

- ✓ Comparison operators
- ✓ Using variables with `print()`

Next Topic

Decision Making 🧠

The next topic will properly cover:

- `True` and `False`
- `if`
- `elif`
- `else`
- `and`
- `or`
- `not`
- Nested `if` statements

🏆 PYTHON BASICS COMPLETE

Python Basics gave me the foundation I need for the next stages of my Python journey.

I have learned how to receive information, store it, convert it, work with different types of data, perform basic calculations, compare values, and create simple interactive programs.

🐍 PYTHON BASICS → ✓ COMPLETE

Next: DECISION MAKING 🧠